

□ SIMBAD et le VO Registry

OBERTO Anaïs – Mars 2025



SIMBAD



□ Services VO pour SIMBAD

- **SIMBAD Authority**

- ivo://cds.simbad|ivo_vor

- **SIMBAD Simple Cone Search**

```
<ri:Resource created="2021-06-12T12:00:00Z"
              status="active"
              updated="2023-06-12T12:00:00Z" ..>
```

```
<identifier>ivo://CDS.SIMBAD/SCS</identifier>
```

```
<capability standardID="ivo://ivoa.net/std/ConeSearch"
xsi:type="cs:ConeSearch">
```

```
  <interface role="std" xsi:type="vs:ParamHTTP">
```

```
    <accessURL use="base">https://simbad.cds.unistra.fr/cone?</accessURL>
```

GAVO Registry



Euro-VO Registry



US-VAO Registry ?

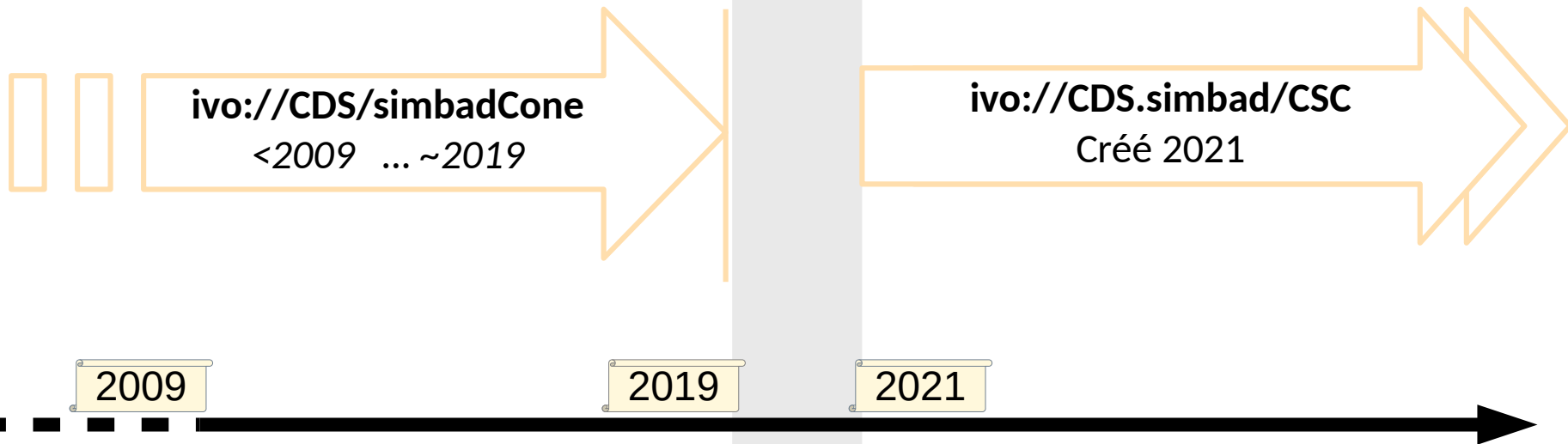
□ Services VO pour SIMBAD

- **SIMBAD TAP query engine**

```
<ri:Resource created="2012-05-10T16:00:00"
              status="active"
              updated="2022-03-25T14:00:00" ..>
<identifier>ivo://CDS.SIMBAD/TAP</identifier>
<capability standardID="ivo://ivoa.net/std/TAP" xsi:type="tr:TableAccess">
  <interface role="std" xsi:type="vs:ParamHTTP">
    <accessURL use="base">http://simbad.cds.unistra.fr/simbad/sim-tap</accessURL>
    <language>
      <name>ADQL</name>
      <version>2.0</version>
```



Historique des déclarations dans les Registries





Historique des déclarations dans les Registries

ivo://CDS/simbadCone
<2009 ... ~2019

ivo://CDS.simbad/CSC
Créé 2021

2009

2019

2021

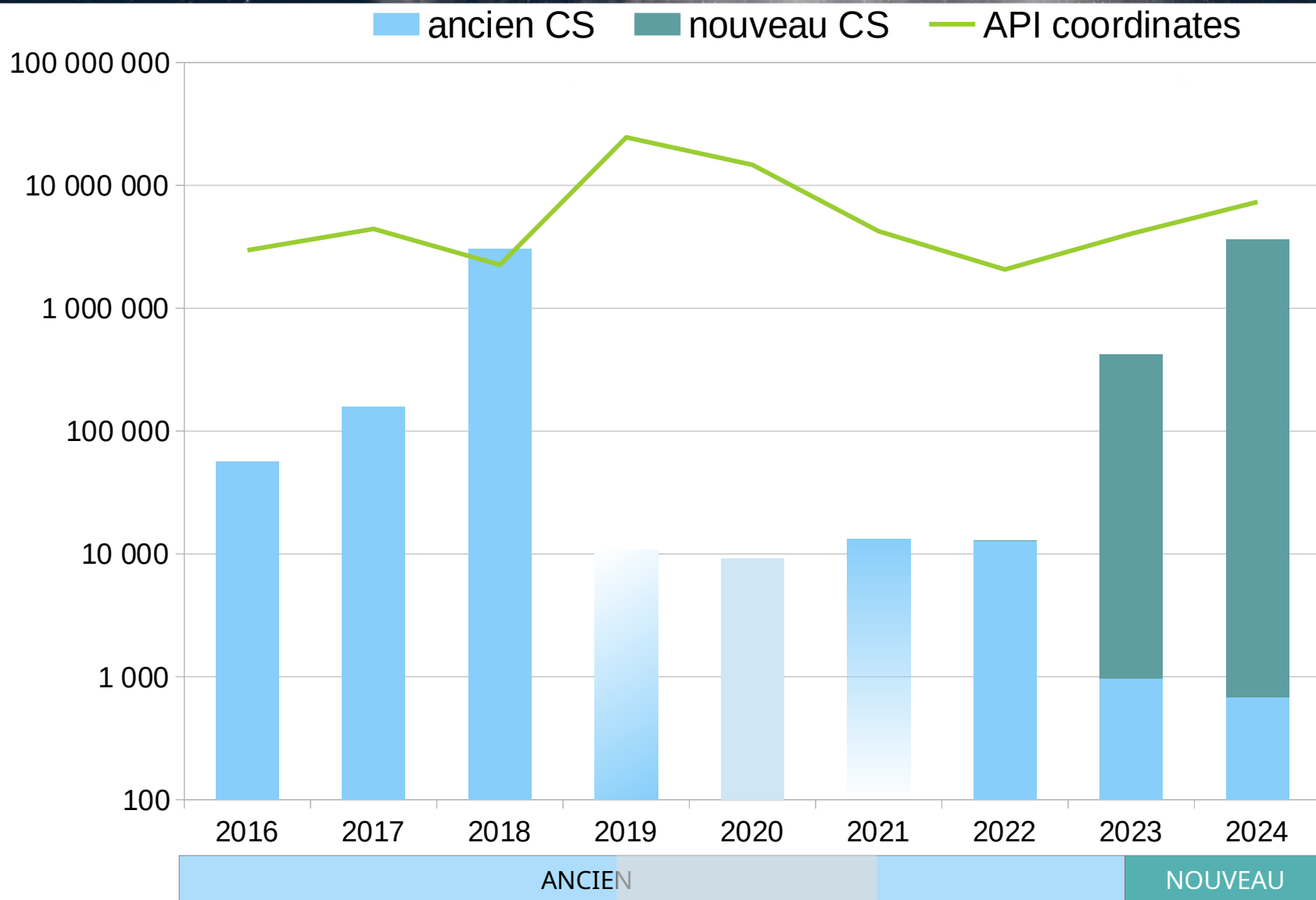
2023

TRES VIELLE
URL

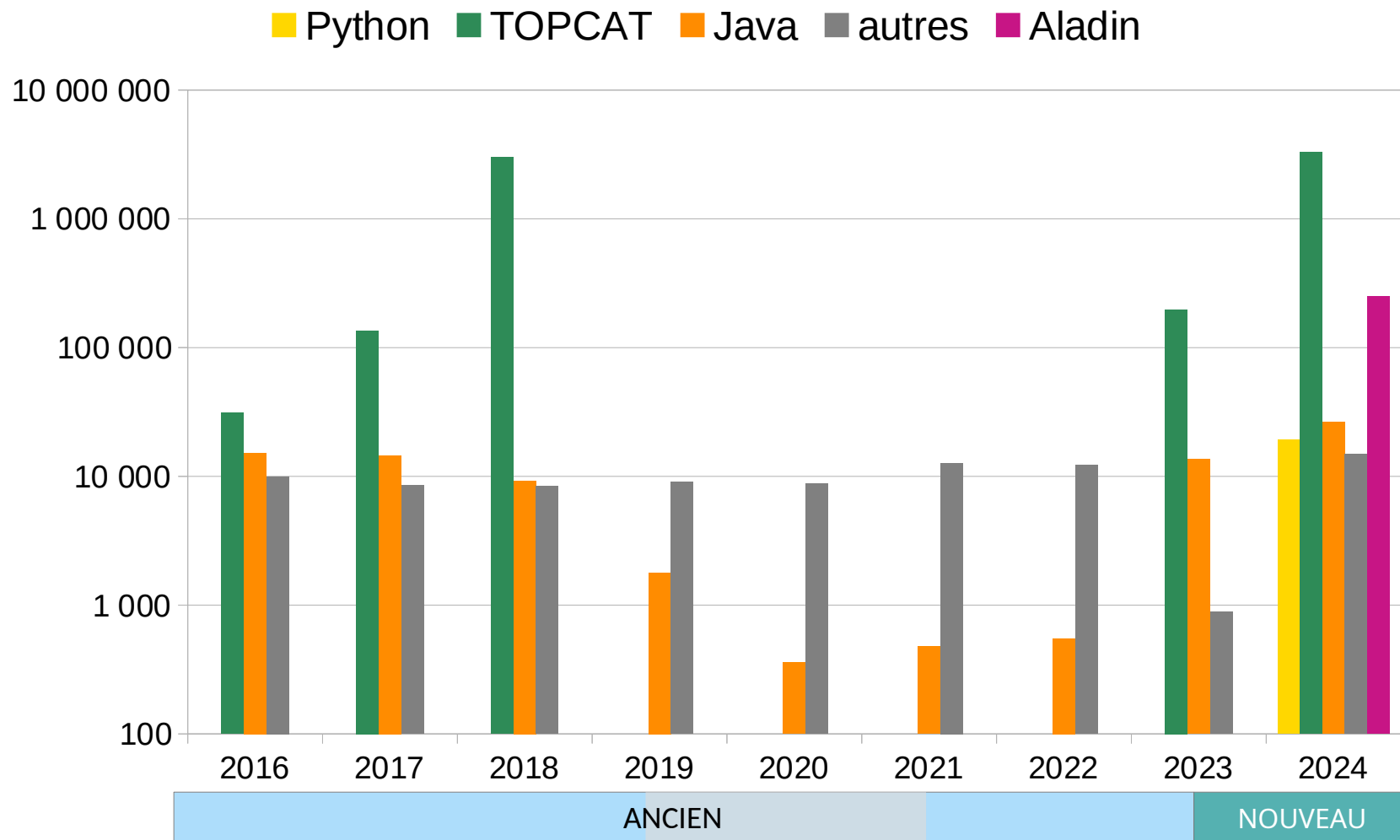
ANCIENNE URL
/sim-cone?

NOUVELLE URL
/cone?

Nombre de requêtes Simbad Cone Search

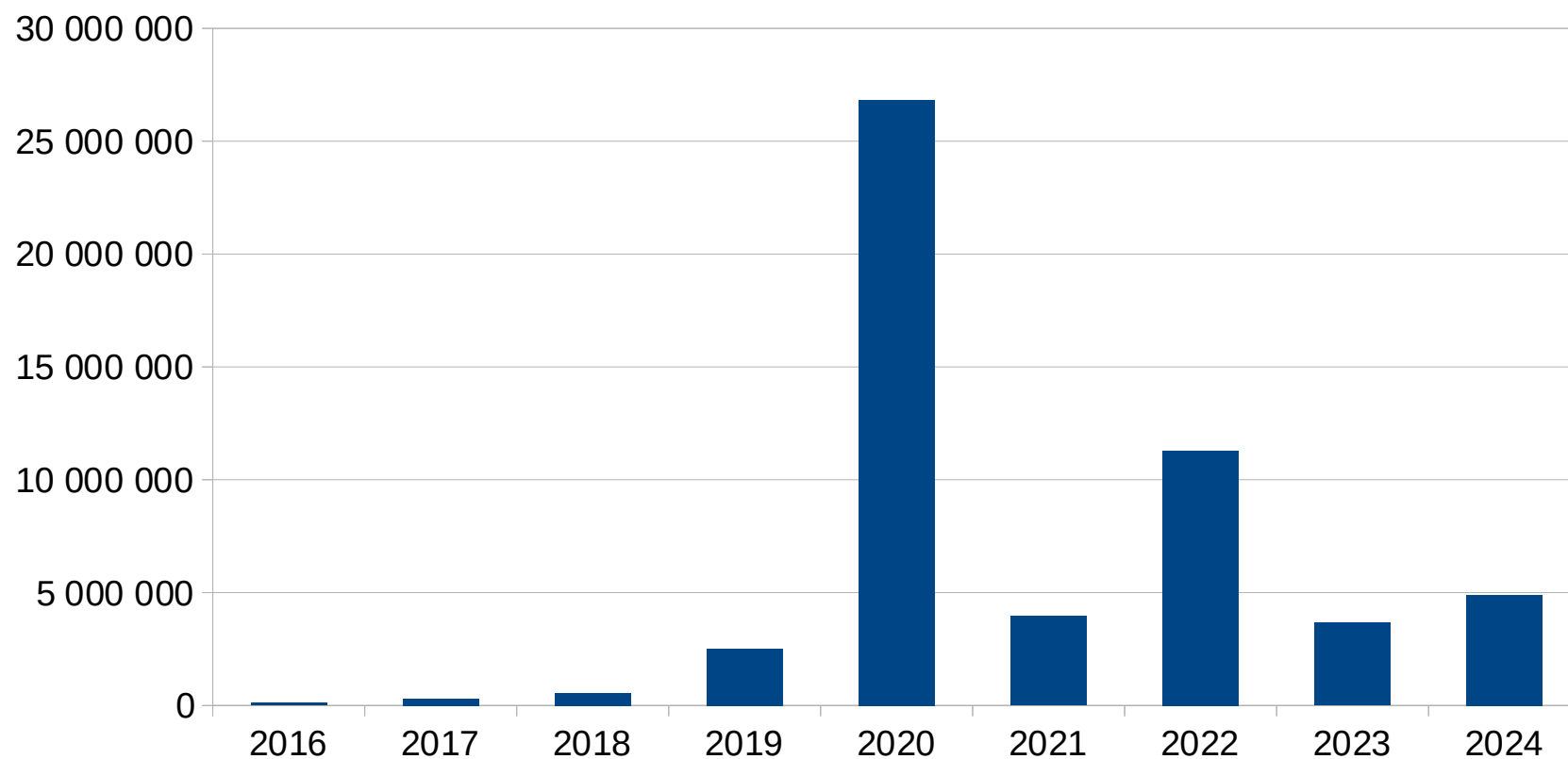


Usage Simbad Cone Search



Usage Simbad TAP

Nombre de requêtes Simbad-TAP par an

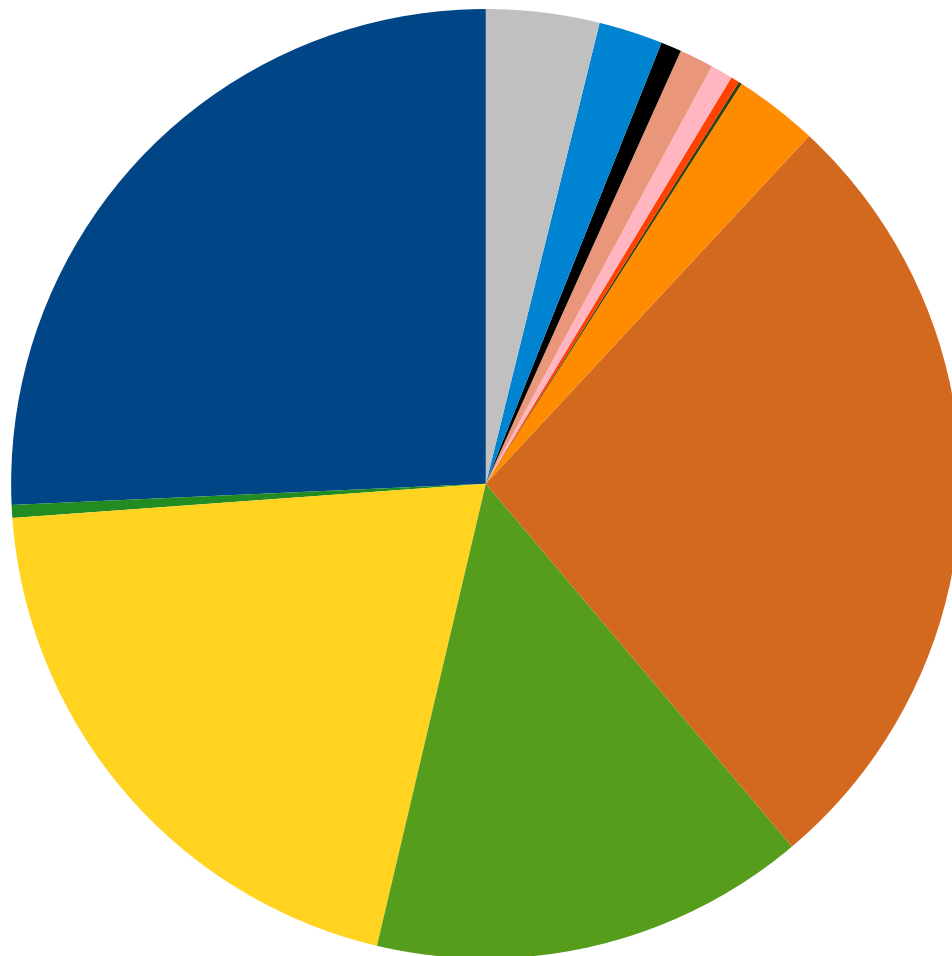


Dans le registry depuis 2012



Origines des requêtes en 2023-2024

- Astrophoto
- STILTS
- Python
- Mozilla
- ALMA
- ESA
- ObsPM
- ObsJapon
- Garching
- Grenoble
- Wolfram
- Autres Java
- Autres



□ Validateurs

VO Monitor - SeaMonkey

File Edit View Go Bookmarks Tools Window Help

Back Forward Reload Stop <http://heasarc.gsfc.nasa.gov/vo/monitor/> Search Print

Home Bookmarks



National Virtual Observatory

Summary and Status of VO Services

NVO Home VO Service Notices NVO Feedback Monitor Help

Hosted by: HEASARC

There are 0 Active Note(s), 0 Pending Note(s), and 0 Total Note(s)

User Tools	Data Services
Portal: 8 out of 8 services are up	Base Tools: SIMBAD: OK MED: OK
Tools: 8 out of 9 services are up	Images: 23 distinct hosts; 45 out of 47 tests passed
	Spectra: 13 distinct hosts; 26 out of 26 tests passed
	Tables: 25 distinct hosts; 23 out of 25 tests passed
	Databases: 27 distinct hosts; 26 out of 27 tests passed

Operations and Testing

Service Tests: 4 distinct hosts; 4 out of 4 tests passed

Publishing Registries: 4 distinct hosts; 4 out of 4 tests passed

Number of Distinct Services Tested: 113
Parallel services at tested sites: 5720

Services Last Tested between: **2009-05-27 20:14:48** and 2009-05-27 21:14:48

We do not check all VO services. Instead, we try to query a representative subset of all services for each VO host. If you feel that an important service is missing from this page please send us a [note](#).

We also provide a tool for finding the status of a service using an IVO identifier: [get_status_help.html](#).

Member Meet the

☐ Validateurs ❤️



- */cone?*
 - ESA-VO : **C** 1 erreur
 - PADC : **3 erreurs**, 4 warnings
- */sim-tap?*
 - ESA-VO : **C--** 4 erreurs
 - PADC : **62 erreurs**, 28 warnings, 1 failure

□ SIMBAD ConeSearch Errors

CONE SEARCH

<https://..simbad../cone?>

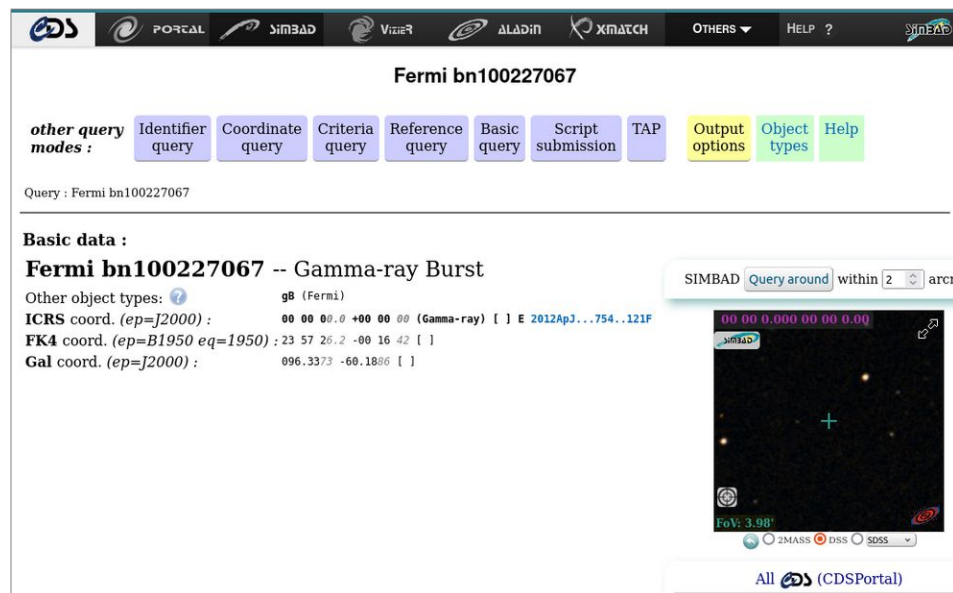
[RA=..&DEC=..&SR=0.1&RESPONSEFORMAT=votable&MAXREC=5&VERB=1](https://..simbad../cone?RA=..&DEC=..&SR=0.1&RESPONSEFORMAT=votable&MAXREC=5&VERB=1)

```
<?xml version="1.0" encoding="utf-8"?>
  <VOTABLE version="1.4" xmlns="http://www.ivoa.net/xml/VOTable/v1.3" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="http://www.ivoa.net/xml/VOTable/v1.3 http://www.ivoa.net/xml/VOTable/votable-
1.4.xsd">
    <RESOURCE name="simbad_cone-search_result" type="results">
      <INFO name="standardID" value="ivo://ivoa.net/std/conesearch">IVOID of the service specification</INFO>
      <INFO name="service_protocol" value="Simple Cone Search 1.03-extended">Used access protocol and
version</INFO>
      <INFO name="publisher" value="CDS/SIMBAD">Data publisher</INFO>
      <INFO name="request" value="https://simbad.cds.unistra.fr/cone/?
RA=0.0&DEC=0.0&SR=0.0&VERB=2&MAXREC=2000&ORDER_BY=distance&ORDER_DIR=ASC&RESPONSEFORMA
T=application%2Fvotable%2Bxml%3Bserialization%3DTABLEDATA%3Bcharset%3DUTF-8">HTTP request URL</INFO>
      <INFO name="request_date" value="2025-03-19T11:18:38.295539Z">Query execution date</INFO>
      <INFO name="contact" value="cds-question@unistra.fr">Publisher email address</INFO>
      <INFO name="QUERY_STATUS" value="OK">Successful query</INFO>
      <COOSYS ID="SIMBAD-COOSYS" system="ICRS" epoch="J2000" equinox="2000"/>
    <TABLE>
      <FIELD datatype="float" name="distance" ucd="pos.angDistance" unit="deg">
        <DESCRIPTION>Distance (in degrees) between this object and the cone-search's target.</DESCRIPTION>
      </FIELD>
      <FIELD ID="main_id" arraysize="*" datatype="char" name="main_id" ucd="meta.id;meta.main">
        <DESCRIPTION>Main identifier for an object</DESCRIPTION>
        <LINK href="https://simbad.cds.unistra.fr/simbad/sim-id?Ident=${main_id}&NbIdent=1"/>
      </FIELD>
      <FIELD datatype="double" name="ra" ref="SIMBAD-COOSYS" ucd="pos.eq.ra;meta.main" unit="deg">
        <DESCRIPTION>Right ascension</DESCRIPTION>
      </FIELD>
```

[...]

SIMBAD ConeSearch Errors

- The **zero response** contains no data
 - The XPATH:
`/votable:VOTABLE/votable:RESOURCE/votable:TABLE/votable:DATA/votable:TABLE
DATA/votable:TR/votable:TD` has been evaluated times: "22 | 0 | 0" .



The screenshot displays the SIMBAD ConeSearch interface. At the top, there is a navigation bar with logos for CDS, PORTAL, SIMBAD, VizieR, ALADIN, XMMATCH, and others. The main title is "Fermi bn100227067". Below the title, there are several tabs for different query modes: Identifier query, Coordinate query, Criteria query, Reference query, Basic query, Script submission, TAP, Output options, Object types, and Help. The "Basic query" tab is selected. The query is "Fermi bn100227067". The results section is titled "Basic data :" and shows the object "Fermi bn100227067 -- Gamma-ray Burst". It lists other object types as "g8 (Fermi)". The ICRS coordinates (epoch J2000) are "00 00 00.0 +00 00 00 (Gamma-ray) [] E 2012ApJ...754..121F". The FK4 coordinates (epoch B1950) are "23 57 26.2 -00 16 42 []". The Galactic coordinates (epoch J2000) are "096.3373 -60.1806 []". On the right side, there is a "SIMBAD Query around" button and a search area with a zoom level of 2 arcmin. A small image of the object is shown with a green crosshair. The bottom of the page has a footer with "All (CDSPortal)".

❑ SIMBAD ConeSearch Errors



- Errors

- Exactly one FIELD must have `ucd="ID_MAIN"`, with an array character type (fixed or variable length), representing an ID string for that record of the table. No match found.
- Exactly one FIELD must have `ucd="POS_EQ_RA_MAIN"`, with type double, representing the right-ascension of the source in the ICRS coordinate system. No match found.
- Exactly one FIELD must have `ucd="POS_EQ_DEC_MAIN"`, with type double, representing the declination of the source in the ICRS coordinate system. No match found.



SIMBAD TAP Errors

Euro-VO Registry



RESULTS #1 x

Search Results (3)

	Short Name	Publisher	Title	Capabilities	Compliance
Q	Simbad	Centre de Donnees astronomiques de Strasbourg	Simbad Authority		
Q	SIMBAD SCS	CDS SIMBAD service	SIMBAD Simple Cone Search	SCS	Partially Compliant
Q	SIMBAD TAP	CDS SIMBAD service	SIMBAD TAP query engine	TAP1.0, Availability, Capabilities, Tables	Partially Compliant

1 of 1 Page size: 100

Displaying 1-3 of 3

SIMBAD TAP

SummaryXMLValidation

Service Access URLs

Service Type	Standard Version	Compliance	Access URL	Valid
TAP1.0	1.0	Partially Compliant (C-)	http://simbad.cds.un.../simbad/sim-ta	11:40 18/03

Validation

			Description	Example query does not val
E_TMNC	REQUIREMENT	Pass		
E_TMNR	REQUIREMENT	Pass		
E_TNTN	REQUIREMENT	Pass	Validation URL	http://simbad.cds.unistra.fr/s
E_TOVO	REQUIREMENT	Pass		
E_TRSV	REQUIREMENT	Pass		
E_TTOO	REQUIREMENT	Pass		
E_TYPX	REQUIREMENT	Pass		
E_UDFE	REQUIREMENT	Pass		
E_UKTB	REQUIREMENT	Pass		
E_UPBD	REQUIREMENT	Pass		
E_UPER	REQUIREMENT	Pass		
E_VOCT	REQUIREMENT	Pass		
E_VTIO	REQUIREMENT	Pass		
E_VVLO	REQUIREMENT	Pass		
E_XID0	REQUIREMENT	Pass		
E_XOFK	REQUIREMENT	Pass		
E_XPRM	REQUIREMENT	Pass		
E_XRS0	REQUIREMENT	Pass		
E_XRS1	REQUIREMENT	Pass		
F_CAIO	extra	Pass		
F_CAPC	extra	Pass		
F_DTIO	extra	Pass		



SIMBAD TAP Errors



IVOA Services Validation

Log in

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[Services](#)

[Errors](#)

[Errors Frequency](#)

[Errors History](#)

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Service Type:* TAP

Min Val Date:* 14 / 03 / 2025

Display

TAP errors

for service url: http://simbad.cds.unistra.fr/simbad/sim-tap

From: jj/mm/aaaa To: jj/mm/aaaa

Show 25 entries

Copy

CSV

PDF

Search:

	Svc	Id	IVOID	URL	Date Val	Type	Num	Name	Message
						error			
	Svc	115218814	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	1	TPJL	(l.757, c.56): cvc-enumeration-valid: Value 'UNKNOWN' is not facet-valid with respect to enumeration 'BOOLEAN, SMAL
	Svc	115218815	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	2	BBUE	(l.757, c.56): cvc-complex-type.2.2: Element 'dataType' must have no element [children], and the value must be valid.

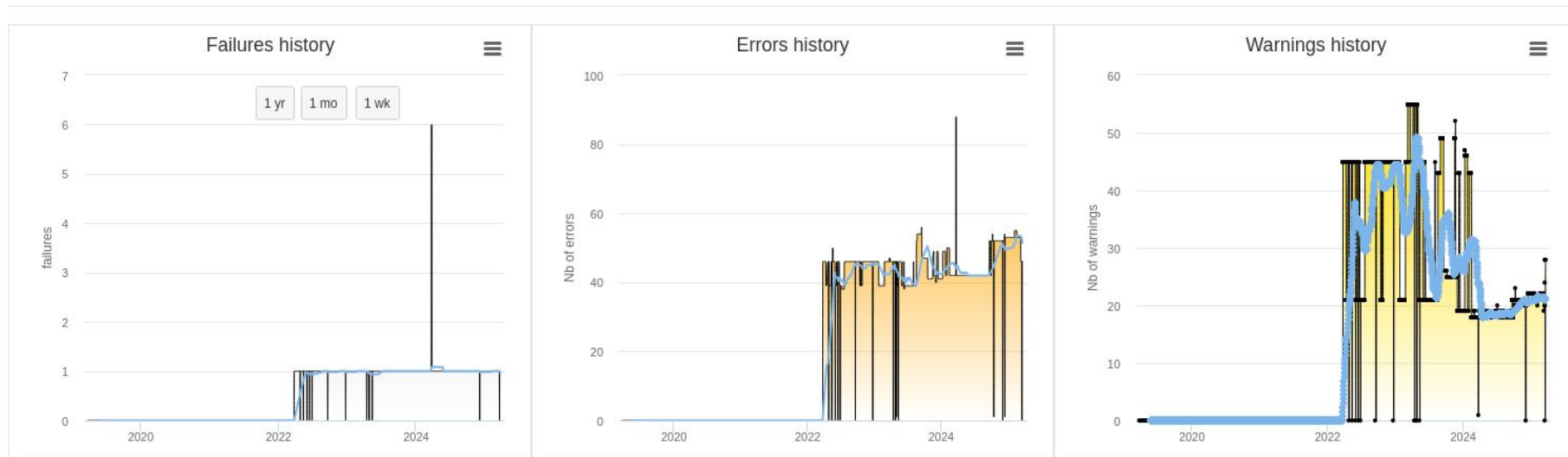
id	115218815
ivoid	ivo://cds.simbad/tap
url	http://simbad.cds.unistra.fr/simbad/sim-tap
svc_type	TAP
date	2025-03-14
type	error
num	2
name	BBUE
msg	(l.757, c.56): cvc-complex-type.2.2: Element 'dataType' must have no element [children], and the value must be valid.
section	TMV

	Svc	115218816	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	3	KGCA	(l.820, c.60): cvc-enumeration-valid: Value 'UNICODECHAR' is not facet-valid with respect to enumeration 'BOOLEAN, S
	Svc	115218817	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	4	BBUE	(l.820, c.60): cvc-complex-type.2.2: Element 'dataType' must have no element [children], and the value must be valid.
	Svc	115218818	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	5	KGCA	(l.830, c.60): cvc-enumeration-valid: Value 'UNICODECHAR' is not facet-valid with respect to enumeration 'BOOLEAN, S
	Svc	115218819	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	6	BBUE	(l.830, c.60): cvc-complex-type.2.2: Element 'dataType' must have no element [children], and the value must be valid.
	Svc	115218820	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	7	KGCA	(l.1077, c.60): cvc-enumeration-valid: Value 'UNICODECHAR' is not facet-valid with respect to enumeration 'BOOLEAN, S
	Svc	115218821	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	8	BBUE	(l.1077, c.60): cvc-complex-type.2.2: Element 'dataType' must have no element [children], and the value must be valid.
	Svc	115218822	ivo://cds.simbad/tap	http://simbad.cds.unistra.fr/simbad/sim-tap	2025-03-14	error	9	KGCA	(l.1569, c.60): cvc-enumeration-valid: Value 'UNICODECHAR' is not facet-valid with respect to enumeration 'BOOLEAN, S

SIMBAD TAP Errors

TAP errors history

for service url: <http://simbad.cds.unistra.fr/simbad/sim-tap>



- Column name **ADQL** reserved word

- [1]Check content of tables metadata from /tables: Column name is ADQL reserved word 'size' in table cat - should delimit like '"size"' x3



- Error reading **examples** document

- [1]Check content of examples document: Error reading examples document at <https://simbad.cds.unistra.fr:443/simbad/sim-tap/examples>



□ SIMBAD TAP Errors

- **Bad UCD :**

- "phot.mag;em.opt.G"; UNKNOWN_WORD: 1 not recognised UCD word: "em.opt.G" (closest: "em.opt.B", "em.opt.I", "em.opt.R", "em.opt.U", "em.opt.V") in column allfluxes.G
- "phot.mag;em.NIR"; UNKNOWN_WORD: 1 not recognised UCD word: "em.NIR" (closest: "em.IR") in 3 columns: allfluxes.F150W, allfluxes.F200W, allfluxes.F444W
- "meta.record;meta.bib"; BAD_SEQUENCE: Too many (2) PRIMARY UCD words: "meta.record", "meta.bib"! Only one is allowed in a UCD.; 1 PRIMARY UCD word not in first position: "meta.bib"! Such words MUST be in first position. in 4 columns: author.oidbibref, biblio.biblio, has_ref.oidbibref, ref.oidbib
x5



International Virtual Observatory Alliance

IVOA Documents



<https://www.ivoa.net/documents/UCD1+/20241218/index.html>



- Types

- [1]Check table query result columns against declared metadata:
Declared/result type mismatch for column update_date in table basic
(date != char) x6
- Value '**UNKNOWN**' is not facet-valid with respect to enumeration '[BOOLEAN, SMALLINT, INTEGER, BIGINT, REAL, DOUBLE, TIMESTAMP, CHAR, VARCHAR, BINARY, VARBINARY, POINT, REGION, CLOB, BLOB]'. It must be a value from the enumeration. x3
- Value '**UNICODECHAR**' is not facet-valid with respect to enumeration '[BOOLEAN, SMALLINT, INTEGER, BIGINT, REAL, DOUBLE, TIMESTAMP, CHAR, VARCHAR, BINARY, VARBINARY, POINT, REGION, CLOB, BLOB]'. It must be a value from the enumeration. x9
- Element '**dataType**' must have no element [children], and the value must be valid. x9



```
<dataType xsi:type="vod:TAPType">UNKNOWN</dataType>
```



volt

<https://github.com/gmantele/volt/>

- **Upload** result column value mismatch

- [1]Make queries with table uploads: Upload result value mismatch for column
d_string(String) in row 0: 0 != 0 *x10*



STILTS

Starlink Tables Infrastructure Library Tool Set

<https://www.star.bris.ac.uk/~mbt/stilts/sun256/taplnt-usage.html>